An Organizational Information-Processing Profile of First Movers

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The present study examines a set of organizational predictors of first moves. More specifically, the study employed organizational information-processing theory to derive a set of hypotheses relating organizational characteristics to the level of first-mover activity. This archival study used a pooled cross section-time series data set of domestic airlines over an 8-year period.

The results in a Tobit analysis indicate that the level of first-mover activity increases positively as boundary spanning and years of education of the top management team increase; and it declines as formalization and years of industry-specific experience of the top management team rise. These and other results are discussed from the perspective of building competitive advantage.

Introduction

The first-mover advantage has attracted a great amount of attention, both among practitioners and researchers, and many research efforts have focused on it. Lieberman and Montgomery (1988), in their review of the literature on the first-mover advantage, argue that first-mover advantages arise from three primary sources: 1) technological leadership; 2) preemption of assets; and 3) buyer switching costs (see also Day, 1988). Alfred Chandler (1977), in his historical analysis of the rise of American industry, demonstrates that in industry after industry the first firm—and sometimes the first two firms—to establish market dominance were able to maintain it over many generations, for example, Sears and Montgomery Ward's in retailing and U.S. Steel and Bethlehem Steel. These firms were able to construct imposing technological and cost barriers that seriously impeded the activities of late entrants. While some of Chandler's firms were not the first to enter their respective industries, many empirical studies have confirmed the advantages associated with being a first mover (Bond and Lean, 1984; Carpenter and Nakamoto, 1986; Urban et al., 1986;

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Glazer, 1985; Robinson, 1988; Day, 1988; Robinson and Fornell, 1985). For example, Robinson (1988) in a study of a broad cross section of mature industrial businesses found that the average market share at maturity for the initial movers or pioneers was 30% and for the late market entrants only 15%.

Almost all of the research on first movers has examined competitive advantage from the perspective of producing and marketing a radically new product or establishing a new industry. Furthermore, none of the research has attempted to explain why firms move first in the marketplace. The current paper differs from past research on first movers in two important respects. First, it examines the first-mover advantage within the more common context of competitive interaction among a large number of rivals. That is, the research focuses on the competitive advantage gained from acting first with specific market actions such as new promotional campaigns, new pricing actions, mergers, opening new territories, and so on. This perspective departs significantly from past research but is arguably the more typical manner by which firms attempt to build competitive advantage (Porter, 1985). Second, the present research looks at the relationship between organizational characteristics of competitors and the tendency for a firm to move first. Very little is known about the organizational characteristics of first movers or the manner in which they should structure themselves to move first (Porter, 1980). Lieberman and Montgomery (1988) specifically cite such studies as being of utmost importance if we are to increase our understanding of first movers. Hence this paper constructs and tests a theory-based model in order to predict the relationships between specific organizational characteristics and the level of first-mover activity within the context of competitive interaction.

Organizational Characteristics: Theory and Hypotheses

To develop an organizational profile of first movers and develop specific hypotheses, we relied heavily on information-processing theory, which seeks to explain behavior by analyzing the information flows in organizations (Knight and McDaniel, 1979). James Thompson (1967) provided an integrated theoretical perspective on organizational information processing, and his ideas were extended and refined by others, particularly Galbraith (1977) and Tushman and Nadler (1978). Further, many empirical researchers have relied on information-processing theory to explain various types of behaviors in organizations, for example, strategy and structure (Burns and Stalker, 1961; Egelhoff, 1982) and group structure and decision-making (Duncan, 1973; Van de Ven et al., 1976). In particular, information-processing theory has been extensively used to explain first moves as they pertain to the diffusion of innovations (Rogers, 1983; Rogers and Shoemaker, 1971.

In this study we used information-processing theory to help identify the organizational characteristics of first movers. More specifically, the focus is on three aspects of information-processing theory: sensory systems, information-processing and analyzing mechanisms, and decision-making selection and retention.

An organization in its interaction with the environment must develop sensory systems that allow it to manage environmental information. Information-processing theory holds that such sensory systems are actualized in large part through the activities of boundary spanners or individuals operating on the peripheries of the

J BUSN RES 1992:25:231–241

organization. Their essential organizational task is to keep track of the external environment and provide information to key organizational members about it (Leifer and Delbecq, 1978). Thomas Allen (1977), for instance, has shown that top-level managers, many of whom lack the time to stay current with pioneering technological research, rely heavily on a select number of highly respected and involved engineers in their company as the major source of information about environment or what is happening in their key specialty areas. These respected engineers are "technological gatekeepers," spending a large portion of their time talking with professional colleagues at other companies and universities, attending professional meetings, and reading technical journals. Such boundary spanners are critical to the organization because they categorize environmental information and package it in such a way that organizational decision makers can assess its importance before committing to a particular course of action.

Some organizations have proportionately more boundary spanners than others. It was our expectation that organizations with extensive boundary-spanning capability would be characterized by a large number of first moves, since they would know what their competitors were doing and would be in a good position to act. Such organizations would also be more likely to identify opportunities and threats before their competitors and, on the basis of this information, act before them. The hypothesis is:

H1: The proportion of boundary spanners is positively related to the level of first-mover activity of organizations.

It is also probable that the level of first-mover activity among organizations will increase when they have effective mechanisms for analyzing and transferring information from boundary spanners to decision makers (information-processing and analyzing mechanisms). Numerous studies have demonstrated the influence that organizational structure has on the organization's effectiveness (Chandler, 1977; Lawrence and Lorsch, 1967). While structure has many dimensions, one that is particularly important for the processing of information is its degree of formalization or the number of organizational levels through which information must pass before the data collected by boundary spanners can be assessed by decision makers.

Previous research has consistently shown that, as formalization or the number of organizational levels rises, so too does the degree to which the information obtained by boundary spanners will be blocked and/or distorted (Aldrich, 1979; Galbraith, 1977). For example, a "gatekeeper" in an research and development unit in an organization may discover a unique business opportunity when he attends a professional meeting, after which he informs a vice-president of research and development who may modify, delay, or block totally the transmission of this information to the next level of the organization. Sometimes the information reaches the decision makers in such a distorted fashion that it is given scant attention. As a result, an organization is less likely to make a first move when encountering an opportunity. Thus the hypothesis is:

H2: Formalization is negatively related to the level of first-mover activity among organizations.

However, even if sensory systems are operating effectively and formalization is not problematic, decision makers can still fail to react to the information. Hence it is important, as information-processing theory does, to analyze the decision-making selection and retention processes of key decision makers.

According to March and Simon (1958), decision makers tend to perceive familiar information more quickly than unfamiliar information. This generalization is important when comparing the informational search activities of decision makers in firms that specialize in certain product-market areas to those of decision makers in firms that do not so specialize and operate in many markets. Essentially, the variety of information that a firm must manage and process in decision making will decrease as product specialization increases (Rumelt, 1974; Miles and Snow, 1978). Decision makers in product-specialized firms will be able to devote more attention to a fewer number of product areas and they will be more familiar with a specific competitor's actions than they will be if they are forced to study many rather than only a few market niches (Miles and Snow, 1978). In contrast, a lack of familiarity with information can easily lead to delays in problem recognition/opportunity and increase the time it takes to undertake search activities, thus inhibiting the level of first-mover activity of organizations. This logic suggests that:

H3: Product specialization is positively related to the level of first-mover activity.

However, decision-making selection and retention systems cost money, and information search can be very expensive (March and Simon, 1958). As absorbed slack or the slack absorbed in the costs of the organization such as overhead, salaries, and other administrative costs increases, the ability of a firm to have an effective decision-making and retention system should decrease, for the firm has proportionately fewer available resources to monitor the external environment closely, thus inhibiting the potential for first-mover activity (Cyert and March, 1963). Further, strategic choices are significantly constrained when resources are low (Chakravarthy, 1982). The hypothesis is:

H4: Absorbed slack is negatively related to the level of first-mover activity.

Another aspect of the decision-making and retention system is the degree to which an organization can control the markets in which it operates, that is, its market power or market share. If a firm has a high market share, it is more likely to obtain better and more useful information about customer needs simply because of the number of relationships it has developed and the sources of information that are potentially available. As a result, a firm's ability to make first moves and preempt the competition should increase as its market share increases. Hence the hypothesis is:

H5: Market share is positively related to the level of first-mover activity.

The final factor in the decision-making and retention area that was of concern in this study is level of education and industry-specific experience of the top management team. Education and experience can easily bias the search for information and thus adversely affect decision making (March and Simon, 1958). Research has indicated that both less-experienced and more-educated managers tend to be more exhaustive in searching out information than their more-experienced and less-educated counterparts (Hambrick and Mason, 1984). As might be expected, educated managers tend to possess a solid base of information that allows them to compete aggressively in the marketplace. In contrast, managers with more industry-specific experience will be typified by less exhaustive information search procedures, and they will frequently rely on programmed procedures and rules that they have employed many times in the past. Although this stance may be acceptable in a highly certain environment that does not change quickly, it will most probably be detrimental in turbulent, changing environments, for example, the highly competitive airline industry. As a result, we would expect less-educated and more-experienced top management teams to be characterized by a low level of first-mover activity, as they would be less able to perceive and seize market opportunities. The hypotheses are:

H6a: The industry-specific experience of the top management team is negatively related to the level of first-mover activity.

H6b: The educational level of the top management team is positively related to the level of first-mover activity.

Method

This paper is part of a larger, programmatic effort that looks at competitive interaction, both first moves and responses to them by other firms in the U. S. domestic airlines industry (see Smith, Grimm et al., 1991; Smith et al., 1989; Chen et al., 1989; Smith, Grimm, Chen, and Gannon, 1989; Smith, Gannon et al., in press; Grimm, Smith, and Gannon 1989). The focus of this paper is the level of first-mover activity for each firm in the sample.

A series of important competitive first moves by airlines over an 8-year period (from January 1979 to December 1986) served as the basis of the present study. These moves included intraindustry mergers and acquisitions, changes in ticket purchase requirements, promotions, market expansions, special introductory fares, price increases, changes in fare plans or structures, introductions of new kinds of airplanes, joint promotions with distributors, vertical integration with feeders, joint promotions with other firms, market withdrawals, service improvements, introductions of "frequent flyer" programs, and price cuts.

Each issue of *Aviation Daily* over these 8 years was the source for identifying these first moves, and a predesigned, structured coding schedule was employed to identify each first move. This method of research has been termed "structured content analysis" (Jauch et al., 1980). For complete details, see Smith, Grimm, et al. 1991.

Aviation Daily is an airline industry journal with a 50-year history, and it includes the most complete and detailed data on competition in the airline industry. Other industry-specific publications such as *Aviation Week and Space Technology* and *Air Transport World* seem to merely repeat information found in *Aviation Daily*. Executives in the airline industry indicated that this journal is the most complete and objective one that is available, and our own industry analysis agreed with this conclusion.

As noted, the unit of analysis is the level of first-mover activity within each year. This level was calculated for each airline, for each year, by summing the number of times it initiated a first move in a specific year. For example, if an airline made five first moves in 1982 and three first moves in 1983, its corresponding first mover activity for each year would be five and three respectively. If a firm did not make a first move in a given year, it received a score of 0. There were 197 observations in this study: 119 observations with no first moves; 32 observations, one first move; 16 observations, two first moves; 13 observations, three first moves; 9 observations, four first moves; 4 observations, five first moves; 2 observations, six first moves; and 2 observations, eight first moves. This measure of the level of first-mover activity became the dependent variable in the present study.

The method for identifying first moves was very rigorous. The procedure began by examining *each* daily issue of *Aviation Daily* beginning with the last day of 1986. A "key word" methodology allowed the identification of competitive responses, and each of these was painstakingly traced back, day by day, over the 8 years, to find the initial reporting of the first move. Key words included "responding to," "reacting to," "following," and so on. For example, if on December 6, 1986, Piedmont opened a new gate and the article noted that this was a response to American Airlines' new gate opening which occurred on November 6, American Airlines would be listed as the first mover and Piedmont as the responder. This was an extremely tedious task involving countless hours. However, it was possible to trace back reliably and accurately over time to identify the first move in each instance. It was also possible to validate over 85% of a random subsample of first moves through other publications.

Although *Aviation Daily* served as the basic source of data for first moves in this study, it was augmented by other data sources so that the information-processing profile of first movers could be constructed. All of the variables in the data set were standardized so that the scales would be the same. In particular, standardization was necessary to produce the boundary-spanning and formalization scales. This standardization (standard Z-score transformation) generated a new value for each variable with a mean of 0 and a standard deviation of 1. The formula for the Z-score transformation is the recorded value minus the mean divided by the standard deviation.

Measures of boundary spanning and specialization were obtained for each airline, for each year, from the *Air Carrier Financial Statistics*. Measures of organizational levels and additional measures of boundary spanning were obtained from the *World Aviation Directory*. Educational level and years of industry-specific experience were available for each airline, for each year, from the *Dun and Bradstreet Corporate Directory*.

More specifically, the amount of boundary spanning was measured by the number of marketing vice-presidents as a percent of total vice-presidents and expenditures for promotion as a percent of sales. These measures represent the interaction between an organization and its external environment at its periphery. The assumption is that a firm's boundary-spanning activities and informational base will increase as the number of marketing vice-presidents and expenditures for promotion as a percent of sales rise. Adams (1976) specifically indicates that marketing and customer relations personnel are critical boundary spanners (see also Miles and Snow, 1978). *World Aviation Directory* was the source for the information on marketing vice-presidents; *Air Carrier Financial Statistics* contained the information on expenditures for promotion as a percent of sales. These two standardized measures were summed to form the single scale for boundary spanning (alpha = .81).

World Aviation Daily provided information on the degree of formalization or the number of organizational levels. Both Jablin (1987) and Bedeian (1984) suggest that the number of organizational levels is a valid measure of formalization. The total number of departments in the organization and the total number of officers in the company were divided by size (measured by total revenue passenger miles for each airline, for each year). These two variables were standardized and combined into a single scale titled "formalization" (alpha = .76). In this instance it was assumed that, as the number of departments and officers in an organization rises, so too will the number of levels or its degree of formalization.

Dun and Bradstreet Corporate Directory listed all corporate executives, and this information became the basis of our definition of the top management team. The CEO and seven other corporate executives were randomly selected for each airline (There were on average 22 corporate airline executives listed for each airline), for each year, and data were collected on each manager's educational level and years of airline industry experience.

Product specialization was measured by the percent of revenues derived from coach travel (as opposed to first class, freight, etc.). *Air Carrier Financial Statistics* provides information on this measure for each airline, for each year.

Air Carrier Financial Statistics was also the source of information for absorbed slack as measured by total expenses as a percent of sales. The assumption is that firms with high expenses as a percent of sales are monitoring the external environment less closely than other firms. This measure was used by Singh (1986).

Market share was the ratio of firm sales to total industry sales. The assumption in this instance is that greater market share increases the amount of information that a firm can obtain and process when deciding whether to make a first move.

Tobit analysis was used to test the hypotheses. This statistical method is appropriate when there are many observations of the dependent variable with a zero value, while others of these observations have the usual continuum of values. The technique, named after economist James Tobin (Tobin's Probit) has been widely applied in the economic literature. For example, Tobit has been used to study determinants of automobile purchasing behavior where there is a discrete or qualitative choice (purchase or no purchase) in conjunction with a quantitative element (amount of purchase). Maddala (1983) provides details regarding Tobit analysis.

Results

Table 1 contains the descriptive statistics for this study (means and standard deviations) while Table 2 includes the results of the Tobit analysis. As shown in Table 2, first-mover activity was significantly related to an increase in boundary spanning and a decrease in formalization. As years of industry experience rose and educational level declined, there was a significant decrease in first-mover activity. All of these relationships were in the direction predicted by the theory and hypotheses derived from them. Further, the direction of the relationship between market share and first-mover activity was positive, although not statistically significant.

However, the hypotheses relating absorbed slack and product specialization to first-mover activity were not confirmed.

Discussion

A majority of the hypotheses put forth in this study were confirmed. However, the relationship between market share and first-mover activity, while positive, was

Table 1. Descriptive Statistics

	Mean	SD
Years of top mgt. team education	16.36	1.26
Years of top mgt. team industry experience	23.36	6.30
Absorbed slack	1.01	0.14
Product specialization	.46	0.40
Market share	0.21	0.16
Boundary spanning[a]	0.18	(.05)
Formalization[a]	0.01	(.95)

[a]Variables were standardized before the scales were produced.

not significant. It is conceivable that the relationship may be curvilinear rather than linear; that is, while we expected first mover activity to increase with market share, it is possible that at some point very dominant firms may find first movership more risky and less economically attractive because they have so much more to lose than their competitors. Consequently, we might expect to find a diminishing impact of market share on first movership at very high levels of market power. To test for this possibility, we reestimated the model, including both market share and market share squared to allow for this possibility. The estimate of market share was .56 ($t = 2.34$) and the estimate of market share squared was $-.21$ ($t = -1.65$). Thus these results do provide some support for the notion that the relationship is curvilinear.

Regarding absorbed slack, it could be that the relationship is discrete or dichotomous; that is, firms in serious financial difficulty may cut back on first-mover activity, but resource availability may not act as a constraint for financially healthy firms. To test for this possibility, we reestimated the model with absorbed slack entering as a dummy variable, with absorbed slack dichotomized at the mean value. The dummy variable coefficient in the resulting reestimated model was $-.67$ ($t = -1.45$). Although the relationship is not significant, there is some indication that firms with less absorbed slack are better able to make first moves.

While there are admittedly many technological, market, and macroeconomic factors that are predictive of first moves, the present study has demonstrated that organizational factors are also of great importance. In fact, it is now possible to construct an organizational profile of first movers. They tend to deemphasize for-

Table 2. Results of the Tobit Analysis

Variable	Estimate	Standard error	T statistic
Constant	−1.59	0.39	−4.04
Market share	0.35	0.29	1.26
Product specialization	−0.26	0.28	−0.93
Absorbed slack	0.54	1.07	0.51
Boundary spanning	1.41	0.44	3.23[b]
Formalization	−3.54	0.84	−4.20[c]
Years industry experience, mgt. team	−0.57	0.25	−2.25[a]
Years education, mgt. team	0.95	0.26	3.65[c]

[a]$p \leq .05$, two-tailed test.
[b]$p \leq .01$, two-tailed test.
[c]$p \leq .001$, two-tailed test.
Standard errors were computed from analytic second derivatives. Number of observations is 197.

malization, stress boundary-spanning activities, possess an ample amount of resources or at least are not experiencing financial difficulty, possess a major but not dominant share of the market, and are led by more-educated and less-experienced top management teams than firms that do not move first.

Recent research has confirmed that both economic and organizational factors are important in predicting firm performance, but that organizational factors are twice as important as economic factors (Hansen and Wernerfelt, 1989). If this is true in the area of firm performance, there is reason to believe that it is also true in the area of predicting first moves.

As suggested in the introductory section of this paper, we are not aware of any research that has looked at the level of first-mover activity, and almost all of the first-mover studies treat first moves as the independent variable rather than the dependent variable. Thus the present study presents a new way of looking at the phenomenon of first-moving.

In addition, although we have touched on the issue of innovation, it is probable that the literature on innovation can be used to supplement or complement the information-processing model, thus allowing the generation of additional hypotheses. The innovation cycles of organizations—from the time that a new idea is introduced into an organization until it results in an actual, marketable product which eventually is replaced by a new product—are in some ways similar to the concept of first-mover activity levels. However, there are obvious differences and both the similarities and differences should allow future researchers to develop new insights and hypotheses.

One body of literature, which we were not able to use because of the archival nature of this study, but which does bear directly on first mover activity, is the psychological profile of the top management team. Some organizations possess top management teams whose members are primarily adaptors; that is, they seek to refine existing product lines. Other organizations possess top management teams whose members are primarily innovators; that is, they seek to introduce radically new products (Rogers, 1983). Kirton (1976) has developed a psychological instrument to measure this aspect of organizational creativity and we would expect that top management teams with a large number of innovators would engage in a high level of first-mover activity. Future research might usefully explore this relationship.

In sum, this study has shown that it is possible to develop an organizational information-processing profile of firms that are frequent first movers. While additional research is needed, the results are sufficiently significant and consistent so that we can accept with reasonable certainty the profile that has been constructed.

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